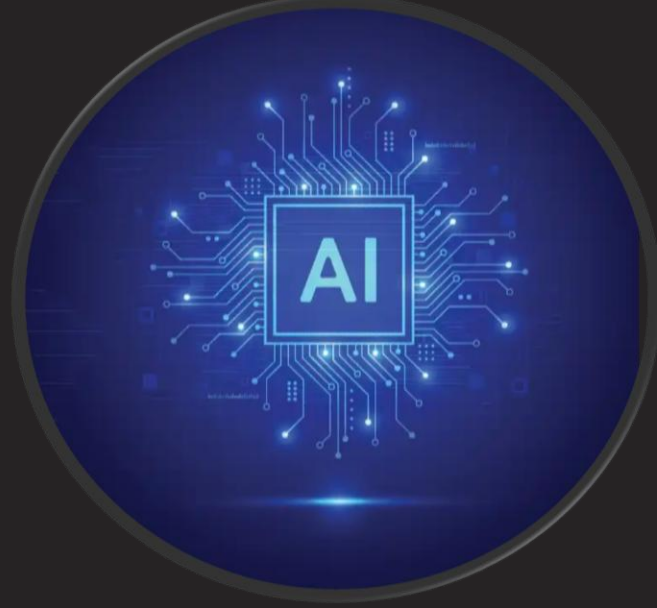




ARTIFICIAL INTELLIGENCE

Artificial Intelligence in Plain Language — and in Depth

*A companion reading to the presentation “Understanding the Tool Behind the Questions,”
prepared for the colloquium on Pope Leo XIV’s encyclical Magnifica Humanitas.*

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How this document is organized

Each idea appears in three layers: a **plain-words recap** of the analogy used in the talk, a **technical explanation** with the proper vocabulary, and a short **worked example**. Words that appear in the closing Glossary are **set in bold** the first time they are used.

You can read it straight through, or jump to any single section — each one stands on its own.

INTRODUCTION

Purpose, and how to use this document

This document is the in-depth companion to a talk given to the Priests of the Sacred Heart in preparation for their discussion of Pope Leo XIV's encyclical *Magnifica Humanitas*. The talk itself was kept deliberately free of jargon, building each explanation on an everyday picture — a kitchen, a child learning the word “cat,” a library, an order placed with a chef. This companion does the opposite. It keeps those same pictures as anchors, but introduces the proper technical vocabulary and develops each idea with examples, so that anyone who wishes to go further can do so at their own pace.

You do not need a technical background to read it. Where a term is unavoidable, it is explained in plain words first and gathered again in the Glossary at the end. Nothing here requires mathematics.

A word on what this document is *not*. It explains the technology only. Where it describes the encyclical (Section IX), it aims to present the document's own argument faithfully and without commentary, and it points you to the official text on vatican.va. The moral and theological weight of these questions belongs to the encyclical itself and to those better placed to interpret it.

A note on accuracy and humility

This field moves quickly, and any single figure or technique may date. The aim here is durable understanding rather than the latest detail: the *shape* of how these systems work changes far more slowly than the names on the products. Where exact numbers appear — costs, dates, sizes — treat them as honest orders of magnitude.

SECTION I

What we mean by “artificial intelligence”

Plain-words recap

In the talk we said it simply: artificial intelligence is a machine doing things that normally take human intelligence — recognizing a face, understanding a sentence, choosing a route. No consciousness, no inner life; only tasks we once believed only people could perform.

Taking the term apart: two words

It helps to separate the two words, because each carries weight. **Intelligence**, in the working sense used here, is a capacity for goal-directed action informed by understanding: to perceive a situation, make sense of it, and decide what to do about it. Perception, reasoning, and action form a single loop — a way of thinking that issues in doing. When a child sees a cup at the table’s edge, grasps that it will fall, and moves it back, that whole arc — notice, understand, act — is intelligence at work.

Artificial marks how the same outward results are produced: by a machine executing **algorithms** — precise, repeatable procedures — that compute the most probable answer from patterns in data. This is the crucial distinction. A person understands the meaning of what they handle; a machine calculates what is statistically likely to come next. The two can yield strikingly similar outputs — a fluent sentence, a correct route — by entirely different means. One rests on comprehension and lived experience; the other on prediction over patterns, without awareness that anything is being understood at all. Keeping this difference in view is what allows an honest conversation about both the power and the limits of these tools.

A more careful definition

More precisely, **artificial intelligence (AI)** is the branch of computer science concerned with building systems that perform tasks associated with human cognition: perception, reasoning, language, planning, and decision-making. The decisive shift of the last two decades is *how* these systems are built. For most of the field’s history, engineers wrote explicit rules by hand. Today the dominant approach is **machine learning**: rather than being told the rules, the system is shown a great many examples and learns the patterns for itself.

A short history, to put “why now” in perspective

1950	Alan Turing asks whether machines can think, and proposes a practical test.
1956	The term “artificial intelligence” is coined at the Dartmouth workshop.
1960s–80s	Rule-based and “expert” systems flourish, then stall — rules cannot cover the world.
1990s–2000s	Statistical machine learning rises; the internet begins to supply data at scale.
2012	Deep neural networks win a landmark image-recognition contest; the modern era begins.

2017 The Transformer architecture is introduced, making powerful language models possible.

2022–today General-purpose chat assistants reach the public; AI becomes an everyday tool.

The science, in other words, is old. What changed recently was not a single invention but a convergence: an ocean of digital **data** to learn from, inexpensive and massively parallel computing power (the same chips built for video games), and the 2017 architectural breakthrough described in Section III.

Two useful ways to classify AI

By breadth of ability.

- **Narrow AI** does one kind of task — translating, filtering spam, recommending a film. Everything in use today is narrow, however impressive.
- **General AI** — a system that could turn its hand to anything a person can, with its own understanding and goals — does not exist; whether it ever will is genuinely debated.
- **Superintelligence**, a system surpassing human ability across the board, is further still into the realm of speculation. When the news says “AI,” it almost always means the narrow kind.

Why this distinction matters for the colloquium

Many fears attach to the *general* and *superintelligent* kinds — the science-fiction machine with a will of its own. The systems actually shaping work, truth, and privacy today are *narrow* tools. Keeping the two apart lets a discussion focus on the real questions in front of us rather than on hypothetical ones.

SECTION II

The family of AI — and the kinds of learning

Plain-words recap: the kitchen

In the talk, the world of AI was a kitchen. The *whole food industry* is artificial intelligence. Inside it, *restaurants that learn their regulars' tastes* are machine learning. Inside those, the *Michelin-starred chef* with a trained palate is deep learning. And the chef who *invents entirely new dishes* is generative AI. Each sits inside the larger one; the analogy shows not just nesting but orchestration.

The technical nesting

These are concentric fields.

- **Machine learning (ML)** is the part of AI in which systems learn from examples.
- **Deep learning** is the part of ML that uses **neural networks** — layered webs of simple units, loosely inspired by neurons — with many layers, allowing the system to build up patterns of increasing abstraction.
- **Generative AI** is the recent branch of deep learning that *produces* new content — text, images, audio, code — rather than only classifying or predicting. A **large language model** is one kind of generative system, specialized in language.

Three kinds of learning

- **Supervised learning** — the system learns from labelled examples (“this email is spam; this one is not”) and generalizes to new cases. Most everyday AI works this way.
- **Unsupervised learning** — the system is given data with no labels and finds structure on its own, such as grouping similar customers together.
- **Reinforcement learning** — the system learns by trial and reward, as one might train a dog: actions that lead to a good outcome are reinforced. This is central to how chat assistants are later refined (Section III).

Worked example: how a child learns the word “cat”

The child-and-cat picture from the talk is, at bottom, a precise illustration of **supervised learning** and of the single idea that makes it powerful: **generalization**. Say “cat” to an infant and nothing happens — the sound is an unlabelled signal with no meaning attached. Show a cat while saying the word, and a *labelled example* is formed: this image goes with this word. One example is fragile. But after a handful — a tabby, a black cat, a kitten — something remarkable occurs: the child correctly names a cat it has never seen before. It has not memorized those particular animals; it has induced the underlying pattern of “cat-ness” and can apply it to new cases.

What the machine does — and where it differs

A model learns the same way in principle: from labelled examples, forming an internal pattern that extends to unseen cases — only from millions of examples rather than a handful, which is both its strength and the source of its appetite for data. Two honest differences remain. The child learned from a real, purring animal it could touch; the machine learned from representations alone — the words and pixels, not the world. And the child needed only a few examples to generalize, where the machine needs an enormous number. Efficiency and grounding, not fluency, are where human and machine learning still part ways.

SECTION III

How a large language model actually works

Plain-words recap: the library of cats

Read one book about cats and you can answer a little; read every book ever written and you can answer almost anything — but you have still never held a cat. That vast reading is **training**; the knowing-words-not-the-world is the limit we return to throughout.

Step 1 — the corpus

Training begins with a **corpus**: an enormous body of text — much of the public web, large collections of books and articles, reference works and code. The scale is measured in *trillions* of words. The quality and balance of this corpus shapes everything the model later does, including its blind spots and biases.

Step 2 — tokens and embeddings

The model does not read letters or whole words but **tokens** — small, common chunks of text (a short word, or a piece of a longer one). Each token is turned into a list of numbers called an **embedding**, a kind of coordinate in a vast “meaning space” where words used in similar ways sit near one another. This is why these systems handle meaning and analogy so fluently: relationships between ideas become distances and directions between points.

Step 3 — pre-training by prediction

In **pre-training**, the model plays one game, billions of times: part of a passage is shown, the next token is hidden, and the model must predict it. Each wrong guess nudges its internal settings — its **parameters**, which number in the billions — very slightly toward a better guess. No facts are typed in by hand. Through sheer repetition, the model absorbs grammar, facts, styles, and patterns of reasoning purely as statistical regularities of language.

Step 4 — the Transformer and “attention”

The architecture that made this possible is the **Transformer** (2017). Its key idea is **attention**: in processing each word, the model weighs how much every other word in the passage should influence it. In “the trophy did not fit in the suitcase because it was too small,” attention is what lets the model link “it” to the suitcase rather than the trophy. Attention also allows the work to be done in parallel, which is what made training at this scale practical.

Step 5 — fine-tuning and human feedback

A freshly pre-trained model is a brilliant mimic but an unruly assistant. Two further steps tame it. **Fine-tuning** continues training on curated examples of helpful, well-formed answers. Then **reinforcement learning from human feedback (RLHF)** has people rank competing answers; the model is rewarded for the kind humans prefer. This is how a raw text-predictor becomes a polite, useful assistant — and where many of its values and refusals are set.

Step 6 — inference, one token at a time

When you use the model, it runs **inference**: given everything so far, it predicts the most likely next token, appends it, and repeats — this is what “**autoregressive**” means. A setting called **temperature** controls how adventurous the choice is: low temperature gives safe, repetitive text; higher temperature gives more variety and surprise. The amount of text it can consider at once — your question plus its working space — is the **context window**.

Worked example: “The cat sat on the ____”

Given “The cat sat on the,” the model assigns probabilities to possible next tokens — *mat* high, *roof* lower, *idea* almost zero — and selects one. It then reconsiders the whole sentence and predicts the next token, and so on. Fluent paragraphs are built this way: not from a plan, but from one good guess after another. This single fact explains both the fluency and the failure mode in Section VII.

The model and the tool

It is worth pausing on a distinction the popular press routinely blurs, because it clears up a great deal. What we have described so far — the trained network of parameters that predicts text — is the **model**. A model is an artifact: in effect a very large file of learned numbers, produced by training, that can be run to generate output. *GPT-4* is a model. It has no screen, no buttons, and no memory of you; it is the trained “brain,” and nothing more. It is our chef, once taught.

What most people actually touch is not the model but a product built around it — an application, often loosely called a **tool**. *ChatGPT* is such a tool. It wraps a model in a conversational interface and typically adds the machinery of the next section: a way to hold the thread of a conversation, to retrieve documents, to call other tools, sometimes to remember across sessions. If the model is the engine, the tool is the car built around it — or, in the language of the talk, the model is the chef and the tool is the restaurant you order from.

The decisive point is that the capability lives in the model, not in the tool. An application is, in large part, an access layer. The very same application can be made dramatically more capable overnight by swapping in a stronger model beneath an unchanged interface; conversely, several different products may sit on top of one shared underlying model. This is why the competition that matters — and the concentration of power the encyclical is concerned with — is fought over models, not over the apps that front them.

Why the distinction matters

Building the model is the hard and expensive part (Section V): the data, the computation, and the months of training. Wrapping a finished model in a usable tool is, by comparison, quick and inexpensive. So the barrier to entry — and the real leverage — rests with whoever controls the model. A practical habit follows: when weighing an AI product, the first question is less “which app is this?” than “which model is underneath it, and who controls that model?”

SECTION IV

Beyond the chatbot — retrieval, tools, and agents

A model on its own knows only what it absorbed during training. It cannot see today’s news, your documents, or your calendar, and it cannot act in the world. Three techniques, now standard in serious systems, address these gaps — and they matter for the colloquium, because they change *what the tool can reach and do*.

Retrieval: giving the model the right pages

Retrieval-augmented generation (RAG) connects a model to a searchable library of trusted documents. When you ask a question, the system first *retrieves* the most relevant passages — using the same “meaning space” of embeddings, stored in a **vector database** — and places them in front of the model before it answers. The effect is a well-read assistant that has just been handed exactly the right pages, so its answer is grounded in your sources rather than in its general memory. This both improves accuracy and lets a model work with private or up-to-date material it never saw in training.

Tools: letting the model call for help

A language model is poor at arithmetic and has no live information, but it can be given **tools** — a calculator, a search engine, a database, a calendar. The model recognizes when a task needs one, issues the request, and weaves the result into its reply. The chef, in the kitchen analogy, learns to pick up the right instrument rather than guessing by hand.

Agents: stringing steps together

An **AI agent** is a model given a goal, a set of tools, and permission to take several steps on its own: plan, act, observe the result, and adjust — repeating until the goal is met. A simple agent might read an inbox, draft replies, and schedule follow-ups. Agents are where today’s most useful — and most consequential — applications are emerging, because a system that can *act*, not only answer, raises sharper questions of oversight and responsibility.

Multimodal models

Early language models handled only text. **Multimodal** models also take in and produce images, audio, and video — describing a photograph, reading a chart, transcribing speech, or generating a picture from a sentence. The underlying principle is unchanged: different kinds of input are turned into the same numeric “meaning space,” so the model can reason across them together.

Why this section matters

Retrieval, tools, and agents are precisely what turn a clever text-generator into something that touches real work, real data, and real decisions. Several of the encyclical’s concerns — oversight, responsibility, the handling of personal information — bear most directly on systems built this way, not on a chatbot answering trivia.

SECTION V

The economics — cost to build, cost to use

Plain-words recap

Building one of these models can cost a fortune; using one, once built, costs pennies. That single gap explains much of both the promise and the worry — the power to create concentrates in very few hands, while the reach extends to nearly everyone.

Why building is so expensive

Training a frontier model means running enormous computations over a trillion-word corpus across thousands of specialized chips for weeks or months. The cost — in hardware, electricity, and expertise — reaches the hundreds of millions. As a result, only a small number of well-funded organizations can train the largest models from scratch. This concentration is itself one of the encyclical's concerns: the common good is poorly served when a decisive technology rests with a few.

Why using is so cheap

Once trained, a model can answer a question for a tiny fraction of a cent, and a single copy can serve millions. This is why capable AI spread worldwide within months rather than decades. The marginal cost of one more answer is negligible — which is wonderful for access, and also what makes misuse cheap and scalable.

Open-weight and closed-weight models

Closed-weight models are kept private and offered as a paid service over the internet; the provider controls them and can update or withdraw them. **Open-weight** models are published for anyone to download, run on their own machines, and adapt. The trade-off is real: open weights bring transparency, independence, and privacy, but also remove the central controls that can prevent misuse. The same openness that lets a hospital run a model privately also lets a bad actor run one without oversight.

Build versus buy — a practical note

Almost no organization needs to *build* a model. The realistic choices are to *use* a provider's model as a service, or to *adapt* an open-weight model with one's own data and guardrails. Most of the meaningful decisions — about privacy, accuracy, and accountability — live in how a model is chosen and surrounded, not in training one.

SECTION VI

Talking to it well — prompting in practice

Plain-words recap

A **prompt** is simply the question or instruction you give the model — the order you place with the chef. A vague order yields an unpredictable plate; a clear one gets you what you actually wanted. The craft of writing good prompts is called **prompt engineering**.

Five habits that reliably help

- **Be specific.** State the task, the form, and the length. “Summarize this in three short bullet points for a non-specialist” beats “summarize this.”
- **Give context.** Tell the model who the answer is for and why; it cannot read your mind or your situation.
- **Show an example.** Paste one example of what “good” looks like and ask for more in the same shape. Supplying examples is called **few-shot prompting**.
- **Assign a role.** “Act as a patient teacher explaining to a beginner” shapes the tone and level of every sentence that follows.
- **Ask for reasoning, for hard problems.** Inviting the model to work “step by step” often improves accuracy on anything involving logic or arithmetic.

Worked example: vague versus precise

Vague: “Write something about our parish event.” → A generic, shapeless paragraph.

Precise: “Write a warm 80-word invitation to our parish summer picnic on 12 July, for families with children, mentioning free food and parking, in a friendly but dignified tone.” → Something you can almost use as-is. The model did not become smarter between the two requests — you simply told it what you wanted.

Two cautions belong here. First, the model will sound equally confident whether it is right or wrong, so verify anything that matters (Section VII). Second, treat a prompt as you would any message that leaves your hands: do not paste confidential or personal information into a public service unless you know how that data is stored and used.

SECTION VII

The honest limits

None of what follows is a reason to dismiss the technology. It is the literacy needed to use it wisely — and much of it connects directly to the encyclical’s concerns.

It can be confidently wrong

Because the model generates plausible-sounding text, it can state falsehoods with complete assurance. Engineers call this **hallucination**. There is no inner sense of truth behind the words — only a sense of what usually comes next. A fluent answer is not a verified one. The practical rule: trust it like a very well-read assistant who is occasionally and convincingly mistaken, and check what matters.

It inherits the biases of its data

A model learns from human writing, and human writing carries human prejudice. Left unchecked, a model can absorb and even amplify **bias** — about gender, race, region, or belief — silently, because the patterns are buried in the data rather than written in any rule. Detecting and reducing this is an active, imperfect discipline.

It has no live knowledge by itself

A bare model knows only its training data, frozen at a past date; it does not know today unless connected to live sources (Section IV). Ask it about recent events and it may answer fluently and wrongly, or simply not know.

Privacy and the appetite for data

These systems are built on data and improved by more of it. That creates a standing pressure to gather, retain, and analyze personal information — conversations, documents, behaviour. The encyclical names **privacy** and the treatment of persons as data among its central concerns; the technical reality is that data is both the fuel and the temptation.

New avenues for misuse

The same fluency that helps can deceive. Convincing fake text, images, audio, and video — **deepfakes** — can now be produced cheaply and at scale, enabling fraud, impersonation, and manipulation of opinion. Models can also be manipulated through their own input (“**prompt injection**”), a security concern wherever a model is connected to tools or private data.

A material footprint

Training and running these systems consumes significant electricity and water for cooling, with real environmental cost. Efficiency is improving quickly, but the footprint is not zero — a consideration that sits naturally alongside the Church’s teaching on care for creation.

SECTION VIII

Reasons to be reassured, reasons to be concerned

Presented side by side, neutrally and without ranking. A balanced view holds both columns at once: the goods are real, and so are the risks.

Reasons to be reassured	Reasons to be concerned
It is a tool, not a conscious being — without will, desire, or awareness.	Its power concentrates with the few who can build it, raising questions of the common good.
It does real good: earlier disease detection, tireless tutoring, accessibility for the disabled.	Whole kinds of work may be displaced faster than people and institutions can adjust.
It lifts dull and dangerous drudgery, freeing human attention for human things.	Convincing fakes and automated persuasion threaten truth and the integrity of public life.
It can widen access to knowledge and expertise once reserved for a few.	Constant data collection erodes privacy and can treat persons as products to be measured.
It remains under human direction: people choose how, and whether, to use it.	Bias learned from data, and autonomous weapons, can cause harm at scale and at speed.

The encyclical’s own list of concerns — job insecurity, manipulation of information, privacy, ideological bias, autonomous weapons, and the vision of an “enhanced” human being — maps closely onto the right-hand column. Its deeper worry reaches beyond any single item, and is taken up next.

SECTION IX

Magnifica Humanitas — a faithful summary

On neutrality

This section reports what the encyclical says; it does not interpret or evaluate it. Paragraph numbers refer to the official text. Any error of summary is the author's, not the document's. The full encyclical is available at vatican.va, and its theology is the proper subject of the session that follows this talk.

What it is

Magnifica Humanitas (“Magnificent Humanity”), subtitled *On Safeguarding the Human Person in the Time of Artificial Intelligence*, is the first encyclical of Pope Leo XIV. It was signed on 15 May 2026 and released on 25 May 2026 — a document of roughly forty-two thousand words in five chapters. It situates the questions raised by AI within the Church's social teaching, in continuity with *Rerum Novarum* (1891) through *Caritas in Veritate* and *Laudato Si'*, and with Pope Francis's recent messages on AI, peace, and communication.

The governing image

The letter opens with a choice: that humanity stands before the decision to build a new tower of Babel, or to build the city where God and humanity dwell together — an image drawn from Saint Augustine's “two cities,” shaped by two loves. This frames the whole document: technology is set not against a backdrop of machinery but of the human heart and what it loves.

The premise: technology is never neutral

The encyclical's load-bearing premise is stated plainly: technology is not a force opposed to humanity (§4), nor is it inherently evil (§9); yet it is never neutral, because it takes on the characteristics of those who devise, finance, regulate, and use it (§16). From this follows the appeal to build for the common good and to “remain human.”

“...technology is never neutral...” — Magnifica Humanitas, §16

The shape of the argument, chapter by chapter

Chapter One — a dynamic approach faithful to the Gospel. The Church's social doctrine is presented not as a fixed handbook but as a living “theology of communion in history” (§27) that, in each age, reads new questions in the light of the Gospel. Artificial intelligence is treated not merely as a new topic but as a development that challenges these categories from within (§17).

The middle chapters develop the foundations: the dignity and grandeur of the human person, and the principles of the common good, subsidiarity, and solidarity — warned to be essential against new monopolies of data and algorithms that would exclude the vulnerable.

Chapter Three — technology and dominance, often called the core of the letter, urges that we not be afraid of present challenges (§91) while honestly naming the dangers of misuse. It takes up the “technocratic paradigm”

— the tendency to let efficiency, control, and profit alone govern decisions (§92) — and recalls that scientific and economic progress, unaccompanied by moral progress, ultimately works against the human person (§94).

The later chapters turn to building civilization through love and preserving humanity through truth, work, and freedom; and finally to peace. On war, the letter warns of AI-accelerated cyberattacks and propaganda (§183) and the distortion of historical memory (§191), and insists that the development and use of AI in warfare be subject to the most rigorous ethical constraints (§197).

The deeper danger it identifies

Beyond the familiar concerns — job insecurity, manipulation, privacy, bias, autonomous weapons, the “enhanced” human being — the encyclical names a danger that underlies them: that human beings may come to see themselves and one another as products. It is this question of what the human person *is*, and must remain, that the document places at its center.

SECTION X

The deeper question

This companion has explained a remarkable tool: how it learns from examples and from an enormous library, how it predicts one token at a time, how it can be extended with retrieval and tools, what it costs, how to use it, and where it fails. Every page has described a *machine*.

The machine can imitate a great deal of what we do — it can write, summarize, translate, converse, and create. And yet, as the child-and-cat showed, it can learn the word without ever meeting the animal: it knows patterns of language, not the world they refer to; it has no body, no experience, no love, no responsibility. That gap is where the human question lives.

When a machine can imitate so much of what we do, what is it that makes the human person irreplaceable?

This is the right question to carry from the technology into the theology. It is posed here, deliberately, and left open — for the encyclical, and for the discussion that follows, to take up.

The glossary on the following pages gathers, in one place, every term set in bold in this document — each with its plain meaning, its technical meaning, and, where one was used, the analogy from the talk.

SECTION XI

Common misconceptions, answered plainly

A great deal of public anxiety and public hype rests on a handful of misunderstandings. Each is stated below as it is commonly heard, followed by what is actually the case. None of these corrections makes the technology less serious — if anything, seeing it clearly is what allows a serious conversation.

“It understands what it is saying.”

In fact: It predicts likely words; fluency is not comprehension. It can describe grief perfectly without feeling anything, because it has learned how the word is used, not what it names.

“It is learning from our conversation as we speak.”

In fact: A model in use is fixed. It does not change from your chat; improvements come only when its makers deliberately train a new version. (Separately, a provider may store conversations to train future models — a privacy question worth checking, but not the same as live learning.)

“It is connected to the internet and knows today’s news.”

In fact: A bare model knows only its training data, frozen at a past date. It can reach live information only when deliberately connected to search or other sources, as described in Section IV.

“If it sounds confident, it is probably right.”

In fact: Confidence and accuracy are unrelated. The same fluent machinery produces true and false statements in exactly the same assured tone; it has no inner signal for which is which.

“It has opinions, feelings, or intentions of its own.”

In fact: It has none. What reads as personality is text shaped during training to be helpful and agreeable. There is no one behind the words.

“Bigger always means better.”

In fact: Size helps, but the quality of the training data, the fine-tuning, and how the model is connected to trusted sources often matter as much or more. A smaller, well-grounded system can beat a larger, unguided one.

“It is objective, because it is a machine.”

In fact: It reflects the data it was trained on and the choices of those who built it. This is precisely the encyclical’s own premise — that technology is never neutral, because it carries the marks of its makers and users.

“It only parrots back what it was trained on; it cannot be creative.”

In fact: It does not store and retrieve passages; it learns patterns and recombines them into genuinely new sentences and images. The results can be strikingly original — which is also why authorship, originality, and credit become harder questions.

“It is just a fancier search engine.”

In fact: A search engine finds and ranks existing pages; a language model composes a fresh answer word by word. That makes it more fluent and more convenient — and also more able to be confidently wrong, since it is writing rather than pointing.

“It is about to become conscious and take over.”

In fact: Today’s systems are narrow tools without will or awareness. A general, self-directed intelligence does not exist, and whether it ever could is genuinely debated. The pressing questions concern how today’s narrow tools are used, by whom, and to what end.

REFERENCE

Glossary of terms

Artificial intelligence (AI)	Machines performing tasks that normally need human intelligence — perception, language, reasoning, choice. <i>Analogy: the whole food industry.</i>
Intelligence	A capacity for goal-directed action informed by understanding: to perceive, make sense, and act. <i>Plainly: a way of thinking that lets you take action.</i>
Artificial	Produced by a machine running algorithms that predict the likely answer, rather than by a mind that understands.
Algorithm	A precise, repeatable procedure a machine follows to turn an input into an output.
Machine learning (ML)	The branch of AI in which systems learn patterns from examples rather than from hand-written rules. <i>Analogy: a restaurant that learns its regulars' tastes.</i>
Supervised learning	Learning from labelled examples so as to handle new, unseen cases. <i>Like a child learning “cat” from a few cats.</i>
Generalization	Applying a pattern learned from examples to cases never seen before — the point of learning.
Deep learning	Machine learning using many-layered neural networks. <i>Analogy: a Michelin-starred chef's trained palate.</i>
Neural network	A layered web of simple units, loosely inspired by the brain, that learns by adjusting its connections.
Generative AI	AI that produces new content — text, images, audio, code. <i>Analogy: the chef who invents new dishes.</i>
Large language model (LLM)	A large generative model specialized in language, trained to predict text. Behind today's chat assistants.
Foundation model	A large, general-purpose model, trained broadly, that can be adapted to many specific tasks rather than only one.
Model	The trained system itself — in effect a large file of learned parameters that generates output. <i>Analogy: the chef, once taught.</i> Example: GPT-4.
Tool / application (product)	The product built around a model that people actually use — an interface, often adding memory, retrieval, and other tools. <i>Analogy: the restaurant.</i> Example: ChatGPT.

Narrow AI	AI that does one kind of task well and cannot step outside it. Everything in use today.
General AI	A hypothetical system able to do anything a person can, with its own understanding. Does not exist.
Training	Teaching a model by exposing it to many examples. <i>Analogy: a child learning from cats; a model, from a whole library.</i>
Corpus	The body of text a model is trained on — often much of the public web, plus books and articles.
Token	A small chunk of text (a short word or part of one) — the unit a model actually reads and predicts.
Embedding	A token or idea as a list of numbers — a coordinate in a “meaning space” where similar ideas sit close.
Parameters	The model’s billions of internal settings, adjusted during training; together they hold what it learned.
Pre-training	The first, largest training stage: predicting hidden next-tokens across the corpus, billions of times.
Transformer	The 2017 architecture, built on “attention,” that made today’s language models possible.
Attention	The mechanism by which a model weighs how much each word should influence its reading of every other word.
Fine-tuning	Further training on curated examples to make a raw model helpful and well-behaved.
RLHF	Reinforcement learning from human feedback: people rank answers, and the model is rewarded for the preferred kind.
Inference	Using a trained model to produce an answer — predicting the next token, appending it, and repeating.
Autoregressive	Generating text one token at a time, each new token conditioned on all that came before.
Temperature	A setting controlling how adventurous the word choices are — low is safe and repetitive, high is varied.
Context window	The amount of text a model can consider at once — your input plus its working space.
Prompt	The question or instruction you give the model. <i>Analogy: the order you place with the chef.</i>

Prompt engineering	The craft of writing prompts that reliably get good results.
Few-shot prompting	Including a few worked examples in the prompt to show the model the shape of a good answer.
RAG	Retrieval-augmented generation: fetching relevant documents and giving them to the model before it answers.
Vector database	A store of embeddings that lets a system find passages by meaning rather than exact words.
AI agent	A model given a goal and tools, allowed to plan and take several steps on its own to reach it.
Multimodal	A model that handles more than text — also images, audio, or video.
Open-/closed-weight	Whether a model is published for anyone to download and run (open) or kept private as a service (closed).
Hallucination	A confident, fluent statement that happens to be false — a consequence of predicting plausible words.
Bias	Systematic unfairness a model absorbs from imbalanced or prejudiced training data.
Deepfake	Convincing fabricated image, audio, or video produced by generative AI.
Prompt injection	A security attack that smuggles hidden instructions into a model’s input to make it misbehave.
Guardrails	The added rules, filters, and checks placed around a model to keep its behaviour safe and on-task.

REFERENCE

Further reading & about the author

The encyclical and its tradition

- ¹ Pope Leo XIV, *Magnifica Humanitas* (2026) — the full text, at vatican.va, with study aids from the Holy See and the USCCB.
- ² Leo XIII, *Rerum Novarum* (1891) — the foundation of modern Catholic social teaching, to which the new encyclical explicitly returns.
- ³ Benedict XVI, *Caritas in Veritate* (2009) — technology as an expression of human freedom, to be governed by charity and truth.
- * Pope Francis, messages on AI, peace, and communication (2024) — the immediate background to the present encyclical.

On the technology, for the general reader

About the author

Dr. Khaled Tannir is an AI and data-at-scale architect based in Montréal (Canada) and a lecturer at McGill University's School of Continuing Studies, where he teaches applied artificial intelligence and large-scale data engineering. He is the author of five technical books on AI and Data at Scale processing and has served as a technical reviewer for many others.

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- ¹ Introductions to artificial intelligence written for non-specialists — chosen for plain language over technical completeness.
 - ² Reputable explainers on how large language models work, and on their limits, from university and public-interest sources.
 - ³ Writing on AI ethics and governance — bias, privacy, transparency, and accountability — for those who wish to go further.

Prepared by Dr. Khaled Tannir as a companion to his presentation. The summary of Magnifica Humanitas in Section IX aims to present the encyclical faithfully and without commentary; any error of summary is the author's and not the encyclical's.

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